IT2011 - Artificial Intelligence and Machine Learning

NetExam-Style Mid Exam Practice: 100 Multiple-Choice Questions

20 questions per lecture. Questions use the observed NetExam pattern: 'Select one or more', five options, close/tricky statements, scenarios, diagram/concept interpretation, and calculation-oriented reasoning.

Coverage: Lecture 01-05 + uploaded Tutorial/Workshop 01-05 + uploaded Lab Sheets 01-04. Lab Sheet 05 was not included because it was not uploaded in the conversation.

Lecture 1 - 20 Questions

1. Which statement best describes Artificial Intelligence (AI) as presented in the lecture?

Select one or more:

A. A field focused only on robotics
B. A broad computer science field that builds systems simulating human intelligence
C. A method limited to statistical prediction
D. A database technology for storing expert rules
E. A programming language for neural networks

2. Which capabilities are explicitly included among AI capabilities in the lecture?

Select one or more:

A. Reasoning
B. Learning
C. Perception
D. Disk formatting
E. Decision-making

3. Why is ENIAC mentioned in the evolution of AI?

Select one or more:

A. It was the first expert system
B. It introduced deep learning
C. It paved the way for programmable machines and computational thinking
D. It invented the Transformer
E. It was the first chatbot

4. Which statements about the 1956 Dartmouth Workshop are correct?

Select one or more:

A. It formally introduced the term Artificial Intelligence
B. It is presented as the official birth of the AI field
C. It focused only on robotics hardware
D. It helped spark work in problem solving, logic and symbolic reasoning
E. It introduced Transformers in 1956

5. Which statement correctly distinguishes traditional programming from an AI/ML-based system?

Select one or more:

A. Traditional programming learns rules from examples
B. AI/ML systems require every rule to be hand-coded
C. Traditional programming uses human-written rules, while ML can infer patterns from data
D. Both always use the same learning process
E. Traditional programming cannot produce outputs

6. A programmer writes explicit IF-ELSE rules to decide whether a number is even or odd. This is an example of:

Select one or more:

A. Unsupervised learning

B. Traditional rule-based programming

C. Reinforcement learning

D. Generative AI

E. Transfer learning

7. Which statements about Machine Learning (ML) are correct according to the lecture?

Select one or more:

A. ML is a subset of AI

B. ML enables computers to learn from data without being explicitly programmed for every task

C. ML includes supervised, unsupervised and reinforcement learning

D. ML can only work with images

E. ML never uses examples

8. A student compares these two prompts: (1) 'Explain artificial intelligence.' (2) 'Explain AI to a 7-year-old using cartoon characters.' Which statements are correct?

Select one or more:

A. Prompt 2 provides more specific context/audience

B. Prompt 2 is more constrained than Prompt 1

C. Prompt quality can affect relevance of generated output

D. Prompt 1 is necessarily always better because it is shorter

E. This is an example of prompt engineering

9. A bank wants to learn patterns from examples labeled 'fraud' and 'not fraud'. Which learning type best fits the described setup?

Select one or more:

A. Supervised learning

B. Unsupervised learning

C. Reinforcement learning

D. Expert-system reasoning

E. Generative modeling

10. Which statements correctly describe Deep Learning (DL)?

Select one or more:

A. DL is a subset of ML

B. DL uses multi-layered neural networks

C. DL can automatically extract features from raw data

D. DL is unrelated to neural networks

E. DL is limited to tabular databases

11. Which architecture-task pairings are supported by the lecture?

Select one or more:

A. CNN - image processing

B. RNN - sequence data such as text or time series

C. Transformer - language models

D. CNN - only payroll calculation

E. Transformer - invented for database normalization

12. Which sequence best represents the relationship among AI, ML and DL?

Select one or more:

A. DL contains AI which contains ML

B. AI contains ML, and ML contains DL

C. ML contains AI, and AI contains DL

D. They are completely unrelated

E. DL is broader than AI

13. Which examples from the lecture are better suited to AI/ML-type approaches because explicit rules are difficult to write for all cases?

Select one or more:

A. Face recognition

B. Spam email classification

C. Weather pattern prediction

D. Printing a fixed tax form

E. Recognizing complex patterns in large datasets

14. A system receives thousands of labeled email examples and learns to classify new emails as spam or not spam. Which description is best?

Select one or more:

A. Human-written rule table only

B. AI/ML-based system learning patterns from data

C. Unlabeled clustering only

D. Database indexing

E. File compression

15. Which statements about Reinforcement Learning are correct according to the lecture material?

Select one or more:

A. It is one of the main types of ML

B. It is associated with learning through rewards

C. It requires every training example to have a correct label

D. The tutorial describes it as reward-based learning used in games and robotics

E. It is identical to supervised learning

16. Which of the following are branches/core areas shown around AI in the lecture?

Select one or more:

A. Machine Learning

B. Deep Learning

C. Expert Systems

D. Natural Language Processing

E. Robotics

17. Which statement about labeled and unlabeled data is most appropriate?

Select one or more:

A. Labeled data is paired with known outputs/labels

B. Unlabeled data has no target labels supplied for the learning task

C. Supervised learning commonly uses labeled data

D. Unlabeled data is always useless

E. Both terms mean exactly the same thing

18. A model is given input-output examples and learns a mapping used to predict on new unseen inputs. What is the central idea?

Select one or more:

A. Traditional hard-coded logic

B. Machine learning generalization from examples

C. Manual database entry

D. Packet routing

E. Disk formatting

19. Which statements are supported by Lab Sheet 01 about traditional programming limitations?

Select one or more:

A. It is effective for simple, well-defined problems

B. It struggles with complex tasks where rules are hard to define

C. Face recognition can be difficult to handle with explicit rules for every case

D. Spam classification and weather prediction can involve intricate patterns

E. Traditional programming automatically learns from labeled examples

20. Which statements about the Lab 01 ML implementation are correct?

Select one or more:

A. scikit-learn provides the LogisticRegression model used for a binary task

B. The fit method trains the model using X and y

C. The predict method applies learned patterns to new inputs

D. numpy is used for numerical array operations

E. Logistic Regression is described as a classifier for binary tasks such as pass/fail

Lecture 2 - 20 Questions

21. Which statement best describes an Expert System?

Select one or more:

A. A system that only stores raw data

B. A program designed to simulate human expert decision-making in a specific domain

C. A neural network that always needs images

D. A spreadsheet with formulas

E. A file compression algorithm

22. Which statements correctly distinguish data, information and knowledge?

Select one or more:

A. Data can be raw facts

B. Information can be organized data

C. Knowledge can represent meaningful insight derived from information

D. Data is always more abstract than knowledge

E. Knowledge is identical to an unprocessed value

23. Which statements about a conventional computer system versus an Expert System are correct?

Select one or more:

A. Conventional systems use data to control processing

B. Expert Systems use encoded knowledge to control reasoning

C. Expert Systems can provide explanations of conclusions

D. Conventional systems always store knowledge separately from control

E. Expert Systems never use rules

24. Which are core components of the Expert System architecture presented in the lecture?

Select one or more:

A. Knowledge Base

B. Inference Engine

C. User Interface

D. Explanation Facility

E. Developer Interface

25. What is the main role of the Knowledge Base?

Select one or more:

A. Store facts and rules collected from domain experts

B. Render charts only

C. Execute operating system commands

D. Compress training data

E. Replace the user interface

26. Which examples fit the types of knowledge shown in the lecture?

Select one or more:

A. Declarative knowledge - facts

B. Procedural knowledge - rules

C. 'Fever is above 100.4 F' - a declarative fact

D. 'IF fever AND sore throat THEN possible flu' - a procedural rule

E. Both types are exactly the same

27. What is the main role of the Inference Engine?

Select one or more:

A. Apply reasoning to the Knowledge Base to derive conclusions

B. Collect user passwords only

C. Replace all domain knowledge

D. Store temporary browser files

E. Format a hard disk

28. Which statements about the Inference Engine are correct?

Select one or more:

A. It uses if-then logic

B. It supports forward chaining

C. It supports backward chaining

D. It is the same thing as the User Interface

E. It derives conclusions from available facts and rules

29. A patient enters fever, headache and runny nose. The system starts from these known facts and applies rules until it reaches a possible diagnosis. Which reasoning approach is this?

Select one or more:

A. Backward chaining

B. Forward chaining

C. Unsupervised clustering

D. Regression

E. Generative decoding

30. A system starts with the goal 'Does the patient have flu?' and then checks which rules could support that goal before asking for required symptoms. Which approach is this?

Select one or more:

A. Forward chaining

B. Backward chaining

C. KNN

D. PCA

E. Gradient descent

31. Which statements correctly compare forward and backward chaining?

Select one or more:

A. Forward chaining is data-driven

B. Backward chaining is goal-driven

C. Forward chaining starts from known facts

D. Backward chaining starts from a suspected goal

E. They are identical reasoning strategies

32. What is Working Memory / Fact Base used for?

Select one or more:

A. Temporarily storing current facts during a reasoning session

B. Permanently replacing the Knowledge Base

C. Storing current symptoms during diagnosis

D. Converting images to vectors only

E. Tracking reasoning progress

33. What is the purpose of the User Interface in an Expert System?

Select one or more:

A. It acts as a communication bridge between user and system

B. It allows users to enter information and receive decisions or suggestions

C. It can be text-based, menu-driven or graphical

D. It stores all expert knowledge instead of the Knowledge Base

E. It is unrelated to user interaction

34. What does the Explanation Facility provide?

Select one or more:

A. The reasoning behind decisions

B. A way to explain how a conclusion was reached

C. Automatic replacement of all rules

D. A user-facing explanation of system reasoning

E. Only raw database records

35. Which statement best describes the Knowledge Acquisition / Developer Interface?

Select one or more:

A. It helps experts and engineers add or update knowledge

B. It prevents any rule changes

C. It is a backstage control area for maintaining knowledge

D. It is the same as the end-user's final answer

E. It can support adding new disease rules in a medical system

36. Which classic Expert Systems and domains are correctly matched in Tutorial 02?

Select one or more:

A. MYCIN - Healthcare

B. DENDRAL - Chemistry

C. XCON - Computing

D. HEARSAY - Military/Voice Recognition

E. MYCIN - Image generation

37. Which statements about the purpose of researching a classic Expert System are correct according to Tutorial 02?

Select one or more:

A. Identify the problem it solved

B. Identify the reasoning method used

C. Identify core components such as Knowledge Base and Inference Engine

D. Describe its real-world impact or benefit

E. Explain how its rules/knowledge were encoded

38. Which statements about ontology are supported by the tutorial material?

Select one or more:

A. An ontology organizes concepts and their relationships

B. It can contain classes

C. It can contain instances

D. It can represent relationships such as isParentOf or hasSibling

E. It is simply a CSV file with no semantics

39. In the family-tree ontology example, which are valid ontology elements?

Select one or more:

A. Class: Person

B. Instance: John

C. Relationship: hasSibling

D. Relationship: isParentOf

E. Only numeric feature vectors

40. Tutorial 02 asks students to compare classic Expert Systems with modern Machine Learning. Which comparison is most appropriate?

Select one or more:

A. Expert Systems can encode domain knowledge as explicit rules

B. ML can learn patterns from data rather than requiring every rule to be manually written

C. The two approaches can differ in how knowledge is obtained

D. Both approaches are identical because both use computers

E. Expert Systems cannot be domain-specific

Lecture 3 - 20 Questions

41. Which sequence best represents the ML project workflow emphasized in the lecture?

Select one or more:

A. Deployment -> Data Collection -> Training -> Problem Definition

B. Problem Definition -> Data Collection -> Preprocessing -> Feature Engineering -> Model Selection -> Training -> Evaluation -> Deployment

C. Data Collection -> Deployment -> Problem Definition -> Evaluation

D. Preprocessing -> Problem Definition -> Deployment -> Training

E. Model Selection -> Data Collection -> Problem Definition -> Deployment

42. What is the main purpose of a Machine Learning Pipeline?

Select one or more:

A. Transform raw data into useful predictions through organized stages

B. Store only final predictions

C. Eliminate the need for evaluation

D. Replace every database

E. Perform only data visualization

43. Which tasks belong to problem definition?

Select one or more:

A. Understanding the domain

B. Identifying the business objective

C. Determining the ML task type

D. Tuning a deployed model only

E. Choosing random_state only

44. Which tasks are part of the EDA workflow in Lab Sheet 02?

Select one or more:

A. Inspect the first and last 5 rows

B. Check data types

C. Find duplicate rows

D. Detect outliers using the IQR method

E. Create visualizations such as scatter plots, histograms and heat maps

45. Which examples represent unstructured data?

Select one or more:

A. Free-form text

B. Images

C. Audio

D. A normalized database table

E. Social media posts and emails

46. Which statements about the EDA and cleaning steps in Lab Sheet 02 are correct?

Select one or more:

A. Irrelevant columns may be removed

B. Column names may be renamed for clarity

C. Duplicate rows can be identified and removed

D. Missing/null values can be handled

E. Box plots can help identify potential outliers

47. Which operations are explicitly part of data preprocessing?

Select one or more:

A. Cleaning

B. Handling missing values

C. Encoding categorical features

D. Normalization/Standardization

E. Final deployment

48. A dataset contains missing temperature readings. Which actions are presented as possible ways to handle missing data?

Select one or more:

A. Fill with the mean

B. Fill with zero or 'unknown'

C. Drop the row

D. Always duplicate the row

E. Ignore missingness without consideration

49. What is the goal of feature engineering?

Select one or more:

A. Transform raw data into meaningful features

B. Improve the ability of a model to learn useful patterns

C. Remove or improve unnecessary information

D. Deploy the final application

E. Replace model evaluation

50. Which statements about feature selection are correct?

Select one or more:

A. Not all information is equally useful

B. Feature selection can reduce irrelevant or redundant features

C. Main categories include Filter, Wrapper and Embedded methods

D. Feature selection is only possible after deployment

E. Model-based methods can use feature importance

51. Which statement best describes Filter Methods?

Select one or more:

A. They select features using statistical tests or intrinsic properties independent of a specific ML model

B. They always require training many models

C. They are generally fast and computationally efficient

D. They are not tailored to one particular model

E. They can be useful for preprocessing large datasets

52. Which statements about Mutual Information are correct?

Select one or more:

A. It measures information shared between a feature and a target

B. It can capture nonlinear as well as linear dependencies

C. Higher mutual information indicates stronger predictive relevance

D. The lecture notes a range from 0 to 1

E. It is only for images

53. What does SelectKBest do?

Select one or more:

A. Selects the top K features based on a scoring function

B. Can use scoring functions such as chi-square, mutual information or ANOVA F-value

C. Can be used as a quick relevance filter

D. Selects every feature automatically

E. Can support classification or regression

54. Which statements describe Embedded Methods?

Select one or more:

A. Feature selection is integrated into model training

B. The model itself can identify important features during learning

C. They are model-specific

D. They never involve a model

E. They can balance performance and computational cost

55. Which statements about scaling in the uploaded labs are correct?

Select one or more:

A. StandardScaler is used in Lab Sheet 03 to standardize numerical features

B. MinMaxScaler is used in Lab Sheet 02 to normalize selected numerical features to 0-1

C. Scaling is a preprocessing operation

D. Categorical handling should be considered separately from numerical scaling

E. Scaling means deleting all non-numeric columns

56. Why is cross-validation used?

Select one or more:

A. A single train/test split can be lucky or unlucky

B. It checks whether performance is consistent across multiple splits

C. It tests the model more than once

D. It removes the need for training

E. It helps assess generalization more reliably

57. Which situation is most consistent with overfitting?

Select one or more:

A. The model learns noise and outliers

B. Training performance is high but test performance is poor

C. The model is too complex

D. The model fails to learn enough from training data

E. The model memorizes training patterns rather than generalizing

58. Which situation is most consistent with underfitting?

Select one or more:

A. The model is too simple

B. It fails to capture important patterns

C. Performance is poor on both training and test data

D. It memorizes noise

E. It does not learn enough from the training data

59. A team trains a model on 80% of the data and tests it on 20%, then deploys only after checking performance. Which statements fit the lecture pipeline?

Select one or more:

A. Train/test split supports evaluation

B. Evaluation should happen before deployment

C. Deployment is a later stage

D. Data preprocessing can be skipped whenever data are messy

E. A good model should generalize to unseen data

60. A dataset has numerical features measured on very different scales (for example, income in thousands and age in years). Which preprocessing decision is most directly relevant?

Select one or more:

A. Consider a scaling method such as normalization or standardization

B. Scaling can make feature magnitudes more comparable

C. Scaling is unrelated to preprocessing

D. Scaling is performed only after deployment

E. The choice of scaling should depend on the nature of the data and model

Lecture 4 - 20 Questions

61. Which statement best describes supervised learning?

Select one or more:

A. Learning from unlabeled data only

B. Training on labeled data where inputs are paired with correct outputs

C. Learning only through rewards

D. Storing expert rules without data

E. Generating images from noise only

62. Which tasks are examples of regression?

Select one or more:

A. Predicting coffee sales as a number

B. Predicting house rent

C. Predicting exam score percentage

D. Predicting spam vs not spam

E. Predicting paddy yield

63. Which tasks are examples of binary classification?

Select one or more:

A. Spam vs Not Spam

B. Malignant vs Benign

C. Pass vs Fail

D. Predicting exact house price

E. Yes vs No

64. Which statements about simple linear regression are correct?

Select one or more:

A. It uses one predictor variable

B. It models a linear relationship with a quantitative response

C. The lecture gives y = beta0 + beta1 x + epsilon

D. It can be used for height and weight relationships

E. It requires a categorical response

65. Which statements about multiple linear regression are correct?

Select one or more:

A. It uses multiple independent variables

B. The lecture gives y = beta0 + beta1 x1 + ... + beta_k x_k + epsilon

C. Each coefficient represents the change in predicted y for a one-unit predictor increase, holding others constant

D. It has exactly one predictor

E. It cannot make numerical predictions

66. Which statement correctly describes the best-fit line in linear regression?

Select one or more:

A. It is chosen so that total prediction error is as small as possible

B. It must pass through every observed point

C. OLS is a common technique for fitting the model

D. It ignores residuals

E. It aims to minimize error

67. For a fitted regression model, what is the residual for an observation?

Select one or more:

A. y + y_hat

B. y - y_hat

C. y_hat - x

D. x - y

E. beta0 + beta1

68. Which quantities are directly involved in the Sum of Squared Errors (SSE)?

Select one or more:

A. Residuals

B. Squared differences between actual and predicted values

C. y_i - y_hat_i

D. Only the predictor x

E. The total of squared prediction errors

69. Which statement about R-squared is correct?

Select one or more:

A. It describes the fraction of variation explained by the estimated model

B. The lecture gives R^2 = (TSS - SSE) / TSS

C. It is unrelated to explained variation

D. A value close to 1 generally indicates a large fraction of variation explained

E. It is the same as the residual for one observation

70. A multiple regression model is y_hat = 10 + 0.65x1 - 1.8x2. Which interpretations are correct?

Select one or more:

A. Increasing x1 by 1 unit changes predicted y by +0.65, holding x2 constant

B. Increasing x2 by 1 unit changes predicted y by -1.8, holding x1 constant

C. The negative beta2 indicates an inverse contribution, all else equal

D. Both coefficients must be positive

E. Beta0 is the intercept

71. Why can logistic regression be used for a binary response?

Select one or more:

A. It models a dichotomous response

B. It uses a sigmoid function to map a score to a probability

C. It can classify outcomes such as 0/1

D. It requires the response to be a continuous salary

E. A common threshold is 0.5 for class assignment

72. For z = beta0 + beta1 x and sigma(z) = 1/(1+e^-z), which statements are correct?

Select one or more:

A. z is a linear score before applying the sigmoid

B. sigma(z) converts the score to a value between 0 and 1

C. With threshold 0.5, P >= 0.5 can be assigned to class 1

D. P < 0.5 can be assigned to class 0

E. The sigmoid is the same as an ordinary least-squares line

73. A logistic model has beta0 = -10 and beta1 = 1.5. At the decision boundary z = 0, which equation is correct?

Select one or more:

A. -10 + 1.5x = 0

B. x = 10/1.5

C. x = 1.5/10

D. The boundary occurs where P = 0.5

E. z = 1 at the boundary

74. Which statements about K-Nearest Neighbors (KNN) are correct?

Select one or more:

A. It selects the closest k data points based on distance

B. For classification, the most frequent class among neighbors can be used

C. For regression, the neighbors' average can be used

D. The lecture/lab uses n_neighbors=5 in an example

E. It never depends on distance

75. For a KNN classifier with k = 5, the nearest five neighbors have classes A, A, B, A, B. What class is predicted?

Select one or more:

A. A

B. B

C. Both equally

D. Cannot be determined from the neighbors

E. Regression

76. Which statements about Decision Trees and Gini impurity are correct according to Tutorial 05 / supervised-learning workshop material?

Select one or more:

A. A Decision Tree can be used for classification

B. Gini impurity can be used to choose a split/root attribute

C. Gini impurity is computed from class probabilities

D. A very deep tree can become problematic through overfitting

E. Decision Trees cannot handle categorical features

77. Which statement about regression trees is supported by the tutorial?

Select one or more:

A. Reduction in Variance can be used to choose good splits

B. They are used only for categorical outputs

C. They can perform regression

D. They cannot split data

E. They are unrelated to decision trees

78. Which statements are supported by Tutorial 04 and Lab Sheet 04 about SVM?

Select one or more:

A. A linear SVM uses a separating hyperplane

B. A decision value can be written as w^T x + b

C. The sign of the decision value can determine the predicted class under a stated rule

D. The lab experiments with linear and RBF kernels

E. SVM requires every real-world dataset to be perfectly linearly separable

79. A classifier has TP=40, TN=50, FP=10, FN=20. Which statements are correct?

Select one or more:

A. Accuracy = (TP+TN)/(TP+TN+FP+FN)

B. Precision = TP/(TP+FP)

C. Recall = TP/(TP+FN)

D. F1-score combines precision and recall

E. Accuracy ignores TN completely

80. Which metrics are explicitly used to evaluate classification models in Lab Sheet 04?

Select one or more:

A. Accuracy

B. Precision

C. Recall

D. F1 score

E. ROC AUC

Lecture 5 - 20 Questions

81. Which statement best describes Generative AI in the lecture?

Select one or more:

A. AI that only classifies existing labels
B. AI that generates brand-new content
C. A database design method
D. A file storage format
E. A rule language

82. Which comparison between AI types is correct?

Select one or more:

A. Discriminative AI focuses on classifying existing data
B. Predictive AI focuses on predicting future/outcome values
C. Generative AI creates new content
D. Generative AI is defined only as spam detection
E. Predictive and generative always mean the same thing

83. Which statements about Neural Networks are correct?

Select one or more:

A. They can have an input layer
B. They can contain hidden layers
C. They can have an output layer
D. Deep neural networks use multiple layers
E. Neural networks are unrelated to deep learning

84. Which statement about the modern language-model architecture in the lecture is correct?

Select one or more:

A. Transformers were introduced in 2017 in the paper 'Attention is All You Need'
B. Modern LLMs use Transformer architecture
C. Transformers are a type of spreadsheet formula
D. The lecture associates Transformers with language models
E. Transformers were introduced in 1946

85. Which models/types are explicitly contrasted in the lecture's language-model section?

Select one or more:

A. Autoregressive models
B. Autoencoding models
C. Language models
D. Only decision trees
E. Only linear regression

86. According to the lecture material, which example is associated with an autoregressive language model?

Select one or more:

A. ChatGPT
B. A decision tree
C. KNN
D. SVM
E. OLS

87. Which definitions are correct according to the lecture?

Select one or more:

A. Foundation model - a large pretrained model for broad general-purpose tasks

B. LLM - a type of foundation model focused on text tasks

C. Multimodal model - handles multiple data types such as text, image, audio and video

D. SLM - a smaller LLM optimized for speed and efficiency

E. LLM means Linear Least-squares Machine

88. Which are examples of Small Language Models listed in the lecture?

Select one or more:

A. TinyLLaMA

B. Mistral 7B

C. Gemma 2B

D. Phi-2

E. ENIAC

89. Which are stated uses/advantages of Small Language Models?

Select one or more:

A. Fast inference

B. Low-resource environments

C. Improved efficiency

D. They must always be larger than LLMs

E. They are optimized for speed

90. Which statements about multimodal models are correct?

Select one or more:

A. They can handle more than one data type

B. Text, image, audio and video are examples of modalities mentioned

C. They are limited to text only

D. The lecture lists GPT-4o, Gemini and Claude 3 as examples

E. Multimodal means multiple input/output data types can be handled

91. Which activity is an example of Generative AI for education or productivity?

Select one or more:

A. Generating a PowerPoint coach

B. Producing a poem from a prompt

C. Creating an AI-generated image

D. Checking whether a number is even using a hard-coded rule

E. Generating a sample email

92. What is Prompt Engineering?

Select one or more:

A. Designing and refining prompts to guide a generative AI model

B. A way to specify desired output and context

C. A hardware manufacturing process

D. It can influence relevance and accuracy of outputs

E. It means deleting the model's training data

93. Which factors are explicitly identified as affecting prompt quality?

Select one or more:

A. Clarity

B. Specificity

C. Tone

D. Constraints such as length or format

E. Screen brightness

94. Which prompt is more specific and likely to produce a focused output?

Select one or more:

A. 'Write about technology.'

B. 'Write a 5-line poem about technology in education for high school students.'

C. 'Do something.'

D. 'Technology?'

E. 'Explain anything you know.'

95. Which statements about Hugging Face in the lecture are correct?

Select one or more:

A. It is an open-source AI/NLP platform/company

B. It hosts pretrained models and datasets

C. It offers tools for text, images, audio and more

D. It requires users to train every model from scratch

E. It can integrate with real-world applications

96. Which statement best distinguishes a foundation model, an LLM and a multimodal model?

Select one or more:

A. A foundation model is broad and pretrained; an LLM specializes in text; a multimodal model handles multiple data types

B. All three definitions are identical

C. An LLM is always smaller than an SLM

D. A multimodal model cannot process text

E. Foundation models are only for spreadsheets

97. Which statement about reasoning models is supported by the lecture?

Select one or more:

A. They are described as LLMs trained with reinforcement learning for complex reasoning

B. They are associated with complex problem solving, coding and multi-step planning

C. They are the same as simple rule-based Expert Systems

D. The lecture mentions reasoning models such as o1 and DeepSeek DeepThink R1

E. They are designed only for image compression

98. Which statements correctly reflect the uploaded Tutorial 05 and its relation to supervised learning?

Select one or more:

A. The tutorial uses Decision Trees

B. It asks students to construct a confusion matrix

C. It includes Accuracy, Precision, Recall and F1-score calculations

D. It includes a Gini impurity calculation

E. It is exclusively about Transformer architecture

99. A student asks an AI model: 'Explain artificial intelligence to a 7-year-old using cartoon characters, in exactly 5 sentences.' Which prompt-engineering elements are present?

Select one or more:

A. Specificity

B. Tone/audience specification

C. Constraints on length

D. A clear task

E. Unlabeled clustering

100. Which statements are consistent with the lecture's view of choosing an approach to solve a problem?

Select one or more:

A. Use existing software when a suitable ready-made solution exists

B. Write your own software when the problem is unique and lacks an existing solution

C. Develop an AI model for problems that require learning from complex data

D. Every problem should be solved by training a new AI model

E. AI is unnecessary for every well-understood problem

Answer Key

Multiple answers may be correct. Check every option independently, following the same logic used in the supplied NetExam screenshots.

1. B
2. A,B,C,E
3. C
4. A,B,D
5. C
6. B
7. A,B,C
8. A,B,C,E
9. A
10. A,B,C
11. A,B,C
12. B
13. A,B,C,E
14. B
15. A,B,D
16. A,B,C,D,E
17. A,B,C
18. B
19. A,B,C,D
20. A,B,C,D,E
21. B
22. A,B,C
23. A,B,C
24. A,B,C,D,E
25. A
26. A,B,C,D
27. A
28. A,B,C,E
29. B
30. B
31. A,B,C,D
32. A,C,E
33. A,B,C
34. A,B,D
35. A,C,E
36. A,B,C,D
37. A,B,C,D,E
38. A,B,C,D
39. A,B,C,D
40. A,B,C
41. B
42. A
43. A,B,C
44. A,B,C,D,E
45. A,B,C,E
46. A,B,C,D,E
47. A,B,C,D
48. A,B,C
49. A,B,C
50. A,B,C,E
51. A,C,D,E
52. A,B,C,D
53. A,B,C,E

54. A,B,C,E
55. A,B,C,D
56. A,B,C,E
57. A,B,C,E
58. A,B,C,E
59. A,B,C,E
60. A,B,E
61. B
62. A,B,C,E
63. A,B,C,E
64. A,B,C,D
65. A,B,C
66. A,C,E
67. B
68. A,B,C,E
69. A,B,D
70. A,B,C,E
71. A,B,C,E
72. A,B,C,D
73. A,B,D
74. A,B,C,D
75. A
76. A,B,C,D
77. A,C
78. A,B,C,D
79. A,B,C,D
80. A,B,C,D,E
81. B
82. A,B,C
83. A,B,C,D
84. A,B,D
85. A,B,C
86. A
87. A,B,C,D
88. A,B,C,D
89. A,B,C,E
90. A,B,D,E
91. A,B,C,E
92. A,B,D
93. A,B,C,D
94. B
95. A,B,C,E
96. A
97. A,B,D
98. A,B,C,D
99. A,B,C,D
100. A,B,C